

Project Title	Spotly: Find your next opportunity
Brief Description	Spotly is a Flutter mobile app (Android + iOS) that helps university students discover jobs, internships, scholarships, fellowships, competitions and research roles ranked by personal fit. It combines AI match scores, skill-based learning paths, a career roadmap, an application tracker and an AI assistant, all backed by Supabase.
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Software Requirement & Design Document

Spotly: Find Your Next Opportunity

1. Preface

Motivation. Students miss scholarships, internships and competitions not because they are unqualified, but because opportunities are scattered across dozens of websites, social media pages and email lists, sorted by recency rather than relevance. Deadlines pass unnoticed, eligibility rules are hard to interpret, and students often do not know which skills to build next. Spotly brings discovery, eligibility matching, skill building and application tracking into one place and uses AI to rank opportunities by fit, not just by date.

Target audience. Undergraduate and graduate university students (primary), especially those in their early years who lack career guidance and first-generation students. Secondary users are university career offices and opportunity providers who publish listings through an admin role.

Document audience. This document is written for the course instructor and for developers, and describes what the system must do and how it is designed.

2. Introduction

Purpose. Spotly is a personalised opportunity and career-growth companion. A student builds a profile (degree, year, GPA, skills, resume) and the app returns ranked opportunities with an explanation of why each one fits, recommended learning modules, a career roadmap and a tracker for everything applied to.

Scope.

- **In scope:** authentication, profile and resume, opportunity discovery with search and filters, AI match score with reasons, learning and skills modules, career role suggestions and roadmap, application tracker, AI assistant chatbot, deadline notifications, admin content management.
- **Out of scope (v1):** submitting applications directly to third-party sites, payments, employer-side recruiting tools, a full web version, video courses hosted in-app.

Objectives.

- Show each student a ranked list of opportunities with a match percentage and plain-language reasons.
- Raise profile completeness and unlock sharper matches by guiding students to add skills and a resume.
- Never miss a deadline: surface upcoming deadlines on the dashboard and send push reminders.
- Provide seamless navigation across Dashboard, Opportunities, Learning, Career, Applications, AI Assistant and Profile with a clean, consistent design.
- Use Supabase for all data storage, sharing and retrieval, and integrate multiple AI features (matching, assistant, insights, resume parsing).

3. Main Features

#	Feature	Description	Tech
1	Authentication	Email/password sign-up and login, password reset, persistent session.	Supabase Auth
2	Personalised Dashboard	Greeting, profile-completeness meter, recommended opportunities, upcoming deadlines, career progress bars and an AI career insight.	RTDB listeners
3	Opportunity Discovery	Search and filter by type (Jobs, Internships, Scholarships, Fellowships, Competitions, Research, Volunteering), mode (Remote/Hybrid/On campus) and deadline; bookmark items.	RTDB
4	AI Match Score	Each opportunity gets a match % with reasons such as 'Your major matches', 'Your year of study qualifies', 'SQL experience recommended'.	Cloud Functions + LLM
5	Learning & Skills	Skill-organised modules by category and level (e.g., REST APIs, SQL, Git) with Tutorial, Docs, Practice and Project resources, difficulty and hours.	RTDB
6	Career Guidance	Roles that fit the student (e.g., Backend Engineer 88%) and a step-by-step roadmap toward the chosen goal.	Cloud Functions + LLM
7	Application Tracker	Track applications with status: Draft, Submitted, Interview, Accepted, Rejected.	RTDB
8	AI Assistant	Chatbot with suggested questions; answers about eligibility, essays and next skills using the student's profile as context.	LLM API
9	Profile & Resume	Edit degree, year, GPA and skills; upload PDF resume; AI extracts skills from it.	Storage, ML Kit, LLM
10	Notifications	Push reminders for closing deadlines and status changes.	FCM

4. Functional Requirements

Priority: H = High, M = Medium, L = Low.

ID	Requirement	Pri.
FR-1	The system shall allow a user to register with email and password and to log in and out using Firebase Authentication.	H
FR-2	The system shall keep the user logged in across restarts and support password reset.	H
FR-3	The user shall be able to create and edit a profile containing name, degree, year of study, GPA, interests, career goal and skills.	H

ID	Requirement	Pri.
FR-4	The user shall be able to upload a PDF resume, which is stored in Firebase Storage and linked to the profile.	M
FR-5	The system shall extract skills from the resume (ML Kit text recognition + LLM) and suggest them to the user for confirmation.	M
FR-6	The system shall calculate and display a profile-completeness percentage and tell the user what to add next.	M
FR-7	The dashboard shall show recommended opportunities, upcoming deadlines, career progress and an AI insight, loaded from Firebase.	H
FR-8	The user shall be able to browse all opportunities, search by keyword and filter by type, mode and deadline.	H
FR-9	The system shall compute a match percentage for each opportunity from major, year, GPA and skills, and rank the list by it.	H
FR-10	The system shall show 'Why this matches you' reasons for every match, generated by the AI module.	H
FR-11	The user shall be able to open an opportunity detail page and save or bookmark it.	H
FR-12	The user shall be able to add an opportunity to the tracker and update its status (Draft, Submitted, Interview, Accepted, Rejected).	H
FR-13	The system shall list learning modules filtered by category and level, and store the user's progress per module.	M
FR-14	The system shall suggest roles that fit the user (with a fit %) and generate a step-by-step roadmap to the chosen goal.	M
FR-15	The user shall be able to chat with the AI assistant, use suggested prompts, and get answers that use the profile as context.	H
FR-16	The system shall send push notifications for deadlines within 3 days and for application status changes.	M
FR-17	An admin shall be able to add, edit and remove opportunities and learning modules and suspend users.	M
FR-18	The app shall provide seamless navigation between all screens using a bottom navigation bar and named routes, preserving tab state.	H

Key data model (Firebase Realtime Database)

Node	Main fields
/users/{uid}	name, email, degree, year, gpa, skills[], interests[], goal, resumeUrl, profileStrength, role, fcmToken
/opportunities/{oppId}	title, organization, type, mode, description, requirements{major, minYear, minGpa, skills[]}, deadline, link, createdBy
/matches/{uid}/{oppId}	score (0-100), reasons[], updatedAt
/applications/{uid}/{appId}	oppId, status (draft/submitted/interview/accepted/rejected), appliedOn, notes
/courses/{courseId}	title, category, level, difficulty, hours, resources{tutorial, docs, practice, project}
/progress/{uid}/{courseId}	hoursDone, completed, updatedAt
/roadmaps/{uid}	goal, steps[{title, status}], roles[{name, fit}]
/chats/{uid}/{msgId}	role (user/ai), text, timestamp

5. Non-Functional Requirements

Category	Requirement
Performance	Cold start under 3 s on a mid-range phone; dashboard loads under 2 s on 4G; match scores for a profile computed and stored within 5 s of a profile change; AI assistant first response within 4 s; lists paginated (20 items per page) and cached locally.
Security	Firebase Auth for identity; Realtime Database and Storage rules so users can read and write only their own profile, resume, matches and applications, and only admins can edit opportunities; HTTPS/TLS everywhere; LLM API keys kept only in Cloud Functions; input validation and file-type checks (PDF, max 5 MB) on resume upload.
Privacy	Only the data needed for matching is collected (GPA, degree, skills); resumes are private to the user; users can delete their account and all data; data is not sold or shared with providers.
Usability	Material 3 design with consistent colours, cards and chips; any main screen reachable in at most 2 taps; clear match badges and progress bars; light and dark mode; accessible font sizes and contrast; helpful empty and error states.
Compatibility	Android 8.0 (API 26) and above; iOS 13 and above; phones and small tablets; single Flutter codebase.
Reliability & Availability	Relies on supabase (99.9% SLA); offline persistence lets users view cached opportunities and tracker; if AI is unavailable, rule-based match scores and default reasons are shown.
Scalability	Serverless Firebase scales with the number of students and listings; indexed queries on type, deadline and match score.
Maintainability	Layered architecture (UI, state, repository, services), Provider/Riverpod state management, unit and widget tests, Git version control.

6. System Architecture

Spotly follows a layered client-serverless architecture. The Flutter app holds the presentation layer and state management. Firebase supplies authentication, real-time data, resume storage, serverless functions and push messaging. AI runs behind Cloud Functions so API keys stay on the server: a match engine (rule-based scoring plus LLM-written reasons), an LLM assistant for chat, insights and essay tips, and a resume parser that combines on-device ML Kit OCR with an LLM.

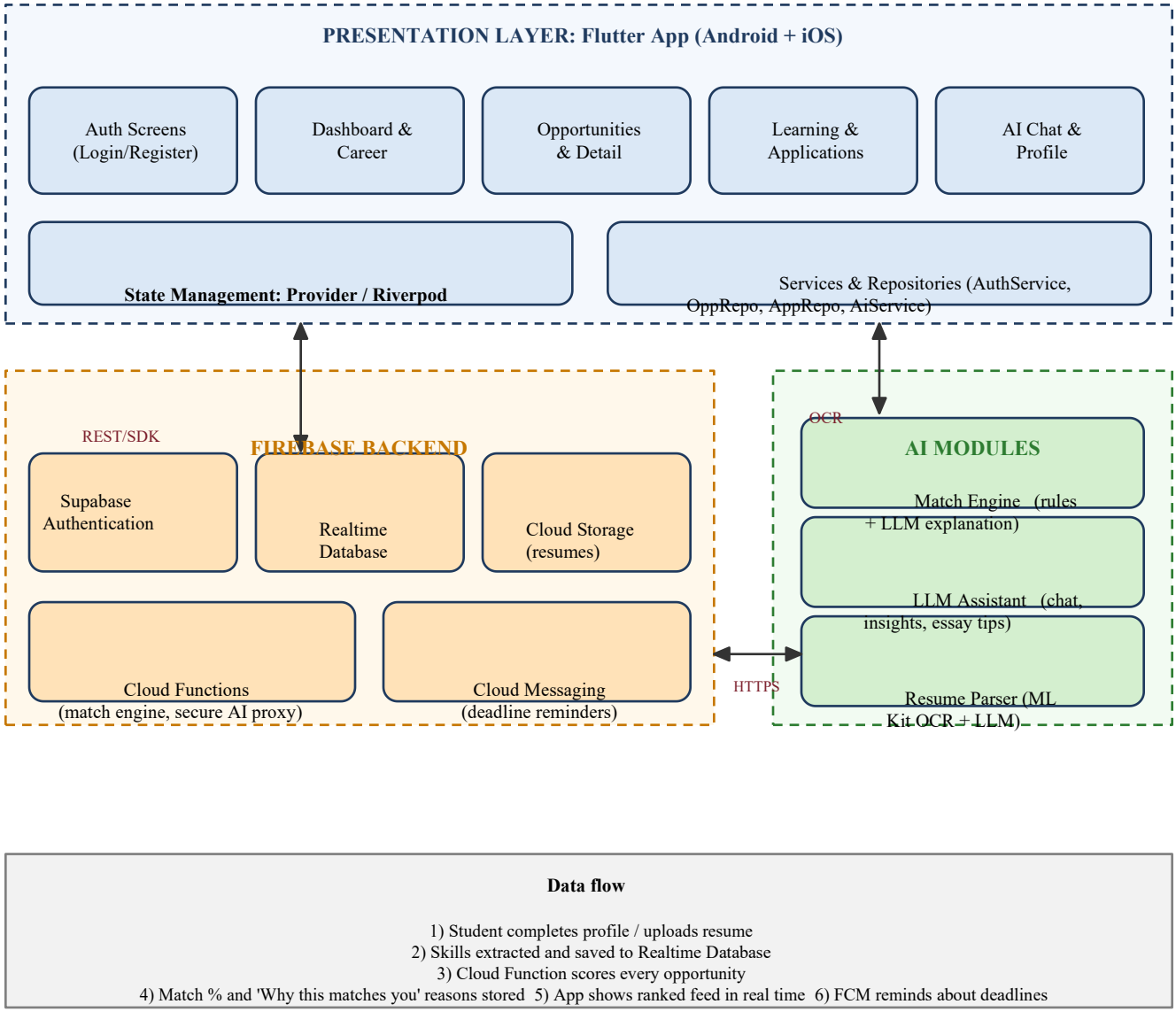


Figure 1: High-level system architecture (front-end, Firebase backend, AI modules)

7. System Models

Figure 2 shows how the actors interact with the system. The Student is the primary actor. The Admin manages opportunities and users. Firebase / AI Services is a secondary system actor (dashed lines) that supports authentication, resume parsing, matching, the assistant and notifications.

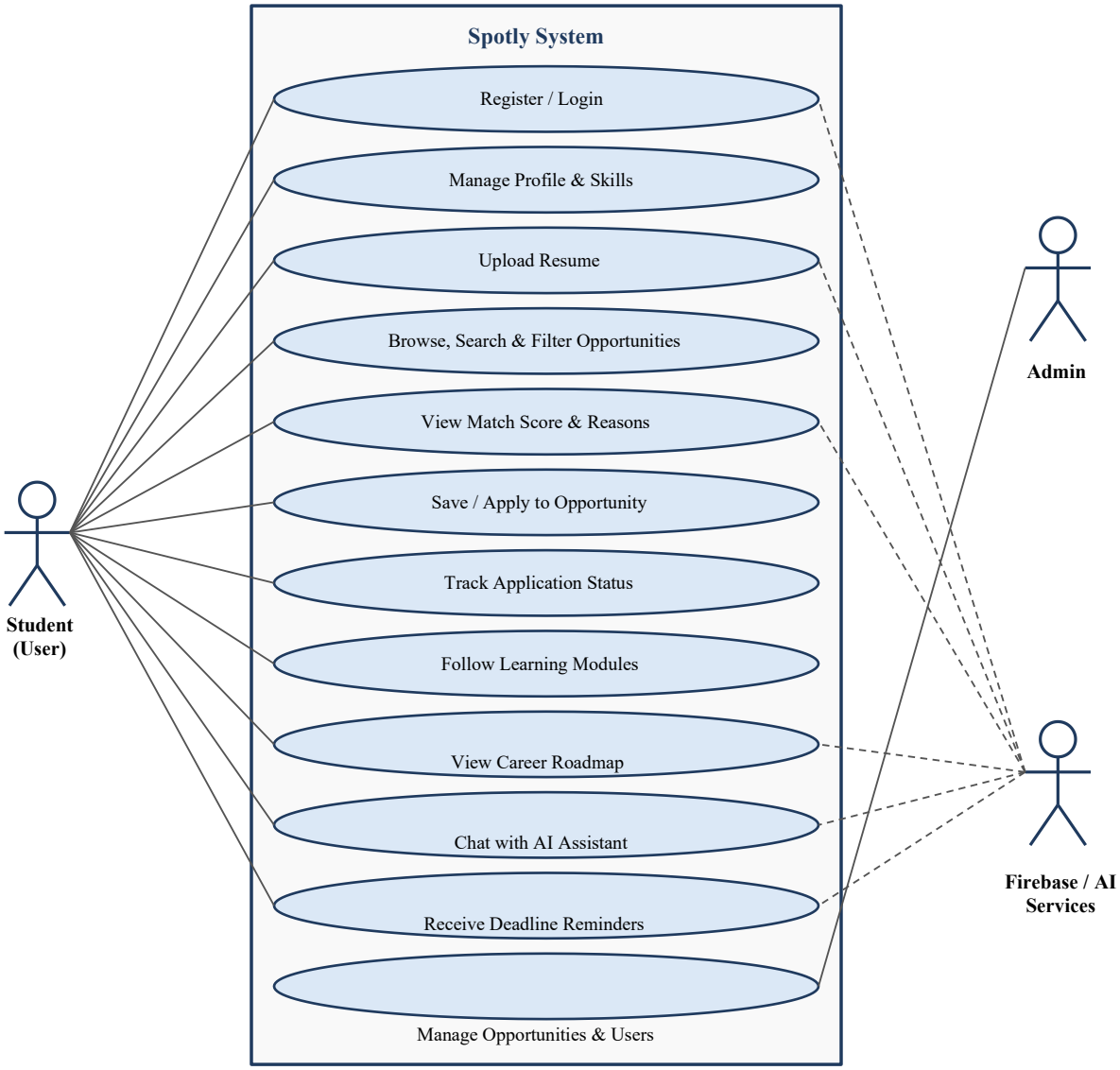


Figure 2: Use case diagram

Use case summary: View Match Score and Reasons

Field	Description
Actor	Student (primary), Supabase / AI Services (secondary)
Pre-condition	User is logged in and has at least a basic profile (degree, year).
Main flow	1) Student opens the Opportunities tab. 2) App reads opportunities and the student's stored matches from Realtime Database. 3) List is shown ranked by match %. 4) Student opens an item and sees reasons such as 'Your major matches'. 5) Student saves it or adds it to the tracker.
Alternate flow	Profile changed: Cloud Function recomputes scores and the list refreshes in real time. AI unavailable: rule-based score with default reasons is shown.
Post-condition	Student has viewed a ranked list; saved items and tracker entries are stored in Firebase.

8. Design and Process Models

Figure 3 is a navigation and process flow model. It shows page-to-page transitions, where Supabase is read or written, and where AI modules are invoked.

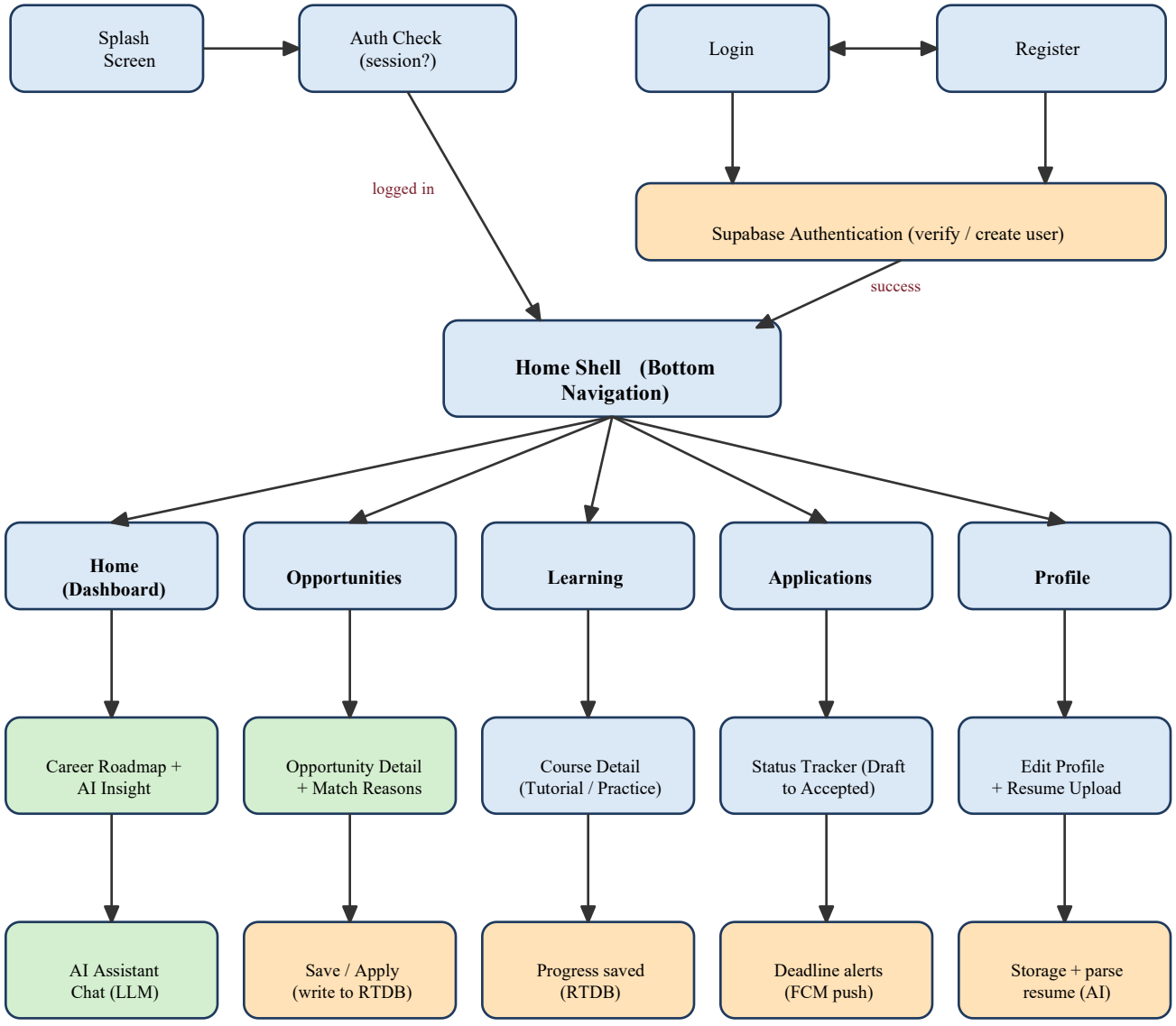


Figure 3: Screen navigation and Supabase / AI interaction flow

Navigation rules

- Splash checks the Supabase session: logged-in users go straight to Home, others to Login.
- Home is a bottom-navigation shell with five tabs (Home, Opportunities, Learning, Applications, Profile); each tab keeps its own navigation stack.
- Career is opened from the Dashboard's career card; the AI Assistant is opened from a floating action button available on every tab.
- Detail pages (Opportunity Detail, Course Detail, Edit Profile) are pushed on top of a tab and return with the back button.

- Logout clears the session and returns to Login, removing all other routes.

9. System Evolution

- **Smarter AI:** embedding-based semantic matching, automatic essay and CV feedback, mock interview practice, and personalised study plans that adapt to progress.
- **More sources:** automatic import of listings from partner websites and university career offices, with duplicate detection.
- **Direct applications:** one-tap apply with auto-filled forms, document vault for transcripts and recommendation letters.
- **Community:** mentor connections, peer groups, success stories and reviews of opportunities.
- **Provider side:** employer and scholarship-provider dashboards, and analytics for university career offices.
- **Platform growth:** Urdu and other languages, web version, calendar integration, and migration to Cloud Firestore for richer queries at scale.

Conclusion. Spotly demonstrates a complete Flutter + Supabase application with clear requirements, a layered architecture, documented models and meaningful AI features, and has a realistic path for growth.